

APIA, Samoa

WMO: 917620

Lat: 13.815S

Lon: 171.781W

Elev: 2

StdP: 101.30

Time Zone: 13.00 (PCS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.0	22.7	18.6	13.4	24.4	19.4	14.2	24.6	11.3	28.0	10.5	27.8	1.1	180	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	5.1	31.2	26.6	30.9	26.5	30.6	26.3	27.4	30.3	27.1	30.1	26.9	29.9	4.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.5	22.0	29.6	26.2	21.6	29.4	25.9	21.3	29.2	86.9	30.6	85.5	30.3	84.4	30.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.4	8.4	19.2	32.4	2.5	1.2	17.5	33.2	16.0	33.9	14.7	34.5	12.9	35.4		
(5)			18.2	28.6	1.7	0.6	17.0	29.0	15.9	29.4	15.0	29.8	13.7	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	27.3	27.5	27.6	27.5	27.2	26.9	26.6	26.4	26.7	26.9	27.1	27.3
	DBStd	0.91	0.92	0.90	0.84	0.72	0.79	0.83	0.89	0.88	0.84	0.76	0.72	0.79
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6237	537	489	546	526	532	508	513	508	501	525	513	537
	CDD18.3	3195	279	256	288	276	274	258	255	249	251	267	263	278
	CDH23.3	32342	2903	2748	3104	2908	2761	2538	2423	2299	2436	2661	2670	2891
CDH26.7	8726	816	811	974	893	773	669	552	502	582	671	698	786	
Wind	WSAvg	3.1	3.0	3.2	2.8	2.5	2.6	3.3	3.5	3.6	3.9	3.3	3.0	2.8
Precipitation	PrecAvg	2971	440	329	357	231	178	151	115	125	164	231	263	370
	PrecMax	4074	960	935	696	528	365	327	292	307	593	579	727	933
	PrecMin	1899	229	79	133	16	10	36	5	19	1	47	25	63
	PrecStd	638	191	154	146	124	81	79	66	79	127	152	156	208
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.5	31.7	31.6	31.6	31.1	30.7	30.2	30.1	30.3	30.7	30.7	31.0
		MCWB	26.5	26.7	26.8	26.8	26.5	26.0	25.5	25.5	25.7	25.9	26.1	26.3
	2%	DB	30.9	30.9	31.1	31.1	30.6	30.2	29.7	29.6	29.9	30.2	30.3	30.5
		MCWB	26.5	26.6	26.8	26.7	26.3	25.8	25.4	25.3	25.5	25.7	25.9	26.2
	5%	DB	30.3	30.5	30.7	30.7	30.2	29.9	29.4	29.3	29.6	29.8	30.0	30.1
		MCWB	26.3	26.4	26.6	26.5	26.1	25.7	25.3	25.1	25.3	25.6	25.8	26.1
	10%	DB	29.8	30.0	30.2	30.3	29.8	29.5	29.1	28.9	29.2	29.4	29.6	29.7
		MCWB	26.1	26.3	26.4	26.3	25.9	25.5	25.1	24.9	25.1	25.4	25.7	25.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.6	27.6	27.7	27.6	27.3	27.0	26.6	26.5	26.6	26.7	26.9	27.3
		MCDB	30.2	30.6	30.5	31.0	30.0	29.9	29.4	29.1	29.7	29.7	29.9	29.9
	2%	WB	27.1	27.1	27.3	27.2	26.8	26.5	26.2	26.0	26.1	26.3	26.5	26.8
		MCDB	30.1	30.1	30.3	30.5	29.9	29.5	29.1	29.0	29.3	29.4	29.6	29.6
	5%	WB	26.7	26.8	27.0	26.8	26.6	26.2	25.8	25.7	25.8	26.0	26.2	26.5
		MCDB	29.6	29.7	30.1	30.1	29.6	29.2	28.8	28.6	28.9	29.2	29.3	29.4
	10%	WB	26.4	26.5	26.6	26.5	26.2	25.9	25.5	25.3	25.5	25.7	25.9	26.2
		MCDB	29.2	29.4	29.7	29.7	29.2	28.9	28.4	28.2	28.5	28.8	29.0	29.0
Mean Daily Temperature Range	5% DB	MDBR	4.7	4.8	5.1	5.4	5.2	5.1	5.0	5.1	4.9	4.9	4.9	4.7
		MCDBR	5.3	5.5	5.6	5.9	5.7	5.5	5.6	5.4	5.4	5.4	5.4	5.1
		MCWBR	2.6	2.7	2.9	3.1	2.9	2.9	2.9	3.0	2.9	3.1	2.8	2.7
	5% WB	MCDBR	4.9	5.0	5.4	5.6	5.4	5.1	5.1	5.1	4.9	5.1	5.0	4.9
		MCWBR	2.6	2.7	2.9	3.1	3.0	3.0	2.9	3.1	2.9	3.1	2.7	2.8
Clear-Sky Solar Irradiance	taub	0.387	0.378	0.375	0.382	0.390	0.391	0.392	0.396	0.399	0.388	0.385	0.385	
	taud	2.625	2.658	2.670	2.643	2.603	2.585	2.573	2.555	2.552	2.596	2.617	2.625	
	Ebn at Noon	956	957	941	902	862	844	851	876	905	937	952	957	
	Edh at Noon	102	98	94	92	90	89	92	99	104	103	102	102	
All-Sky Solar Radiation	RadAvg	5.71	5.71	5.72	5.15	4.54	4.34	4.49	5.20	5.82	6.00	5.88	5.81	
	RadStd	0.57	0.53	0.42	0.42	0.32	0.22	0.18	0.25	0.33	0.30	0.31	0.36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page